GreenLCC India: User Guide

How to install, run and use the tool, screen by screen. Version 2 · 23 Oct 2026

GreenLCC finds the cheapest way, **over the whole life of a building**, to reach an IGBC green rating (Certified, Silver, Gold or Platinum), and shows how safe that answer is if electricity prices and interest rates change. This guide is for anyone using it: the team, the professor, or a building owner. For what the tool is and the input files it reads, see the companion document **GreenLCC_About.pdf**.

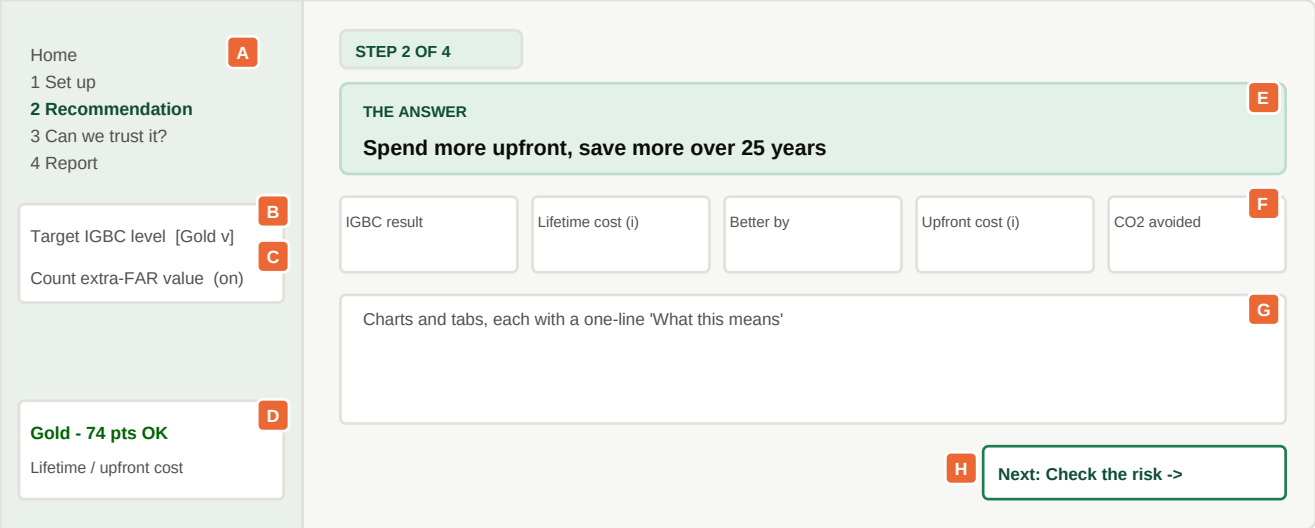
Section	Read it when you want to
1 Install and start	get the app running on your computer
2 The screen at a glance	know what every part of the screen does
3 Screen by screen	know what each page shows and what to click
4 A five-minute demo	present the tool, step by step
5 Using your own project	replace the demo with a real building
6 Reading the results	understand the numbers and words on screen
7 Problems and fixes	something looks wrong

1 Install and start

- You need **Python 3.10 or newer**. Open a terminal in the project folder (the one with app.py).
- Install the libraries once: **pip install -r requirements.txt**. If pip complains about pyarrow and NumPy 1.x, run **pip install "pyarrow<22"**.
- Start the app: **python -m streamlit run app.py**. Your browser opens at **http://localhost:8501**. Keep the terminal open while you use it; press Ctrl+C there to stop.
- Optional: set an **ANTHROPIC_API_KEY** environment variable to let Claude read vendor quotes and polish the memo. Without it, a built-in reader and a template do the same jobs, so nothing breaks.
- Optional check: **python -m pytest tests -q** runs the 60 automated tests (maths, IGBC scoring, optimiser, risk, exports, every screen).

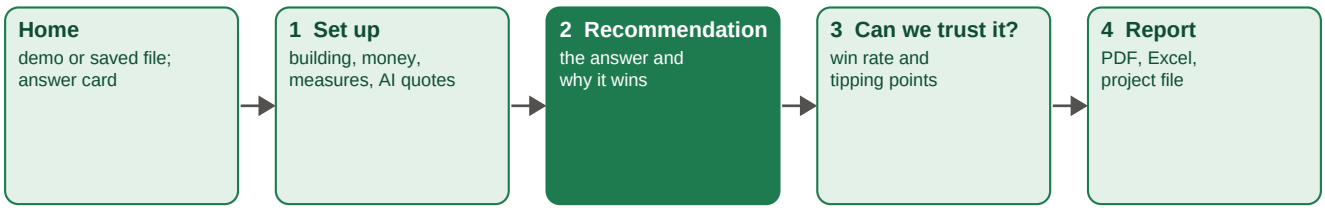
2 The screen at a glance

Every page has the same layout. The letters match the picture below.



	Part	What it does
A	Menu	Home and the four steps. You can jump anywhere, but the steps are meant to be read in order.
B	Target IGBC level	The rating you are aiming for. It applies to every page and to the report.
C	Count extra-FAR value	Some states give extra buildable floor area for a rating. When on, its value is subtracted from lifetime cost.
D	Status box	Always shows the level reached, points, lifetime cost and upfront cost of the recommendation.
E	The answer	One sentence at the top of each page that says what the page found.
F	Number cards	Key figures. Hover the (i) to see the textbook term and how it is calculated.
G	Charts and tabs	The detail. Each chart has a short 'What this means' line under it.
H	Next button	Takes you to the next step, so you never need the menu during a demo.

3 Screen by screen



Home

A short pitch, two ways to start, and why the tool exists.

- Click **Try the Pune demo** to load a ready-made office building, or use **Open a saved project** to load a file you saved earlier.
- As soon as a project is loaded, the **answer card** shows the headline result. Click **See why** to go straight to the recommendation.

1 Set up

Three tabs holding every assumption.

- **Building & money:** plot, floor and roof areas, yearly energy and water use, the electricity tariff and how fast it rises, the discount rate, solar yield, and the FAR bonus values. Change anything and press **Apply changes**. Wrong values are explained, never silently accepted.
- **Measures:** the menu of green options (solar sizes, air-conditioning types, glazing, lighting, water fixtures...). Edit cells in the table, then press **Apply measure changes**. Expand **Replacement parts** to edit what must be re-bought and when; **IGBC credit tables** shows how points are earned.
- **Import a vendor quote (AI):** pick a sample quote or upload a PDF, press **Extract fields**, check the draft (price, maintenance, parts), fill in the engineering inputs a quote doesn't state (energy saved, credit), then **Add to library**. Nothing is added without your review.

2 Recommendation (the main screen)

The answer, the two routes, and the evidence.

- Read the **answer** at the top: how much more you spend upfront and how much you save over the building's life.
- Compare the two route cards: **Cheapest upfront** (what most projects do today) and **Recommended** (lowest lifetime cost). The bar chart shows where the money goes: equipment, maintenance, replacements, leftover value and savings.
- Use **Quick what-if**: drag electricity price growth or the discount rate and see instantly whether the recommendation still wins.
- Tabs: **Other options** (the 10 best mixes), **Points vs cost**, **Credits earned**, **Upgrade path** (cost of going up a level), **Compare measures** (e.g. VRF vs chiller, with payback and carbon), and **How it's calculated**.
- If the target can't be reached, the page says how many points are missing and offers a **Switch target** button.

3 Can we trust it?

How robust the answer is.

- The answer line gives the **win rate**: in how many of thousands of simulated futures the recommendation is cheaper.
- **Tipping points** show the discount rate, tariff growth, tariff level and equipment price at which the ranking would flip. The further from today's value, the safer.
- Tabs: **Simulated futures** (spread of outcomes for both routes), **What matters most** (which assumption moves the answer most), **Interest-rate sweep**. Simulation runs and the random seed are under **Advanced**.

4 Report

Three downloads and the owner memo.

- **Decision report (PDF, 6 pages)**: the document to send. It has the answer, the two routes, the IGBC credit checklist with documents to submit, what to buy, the risk check, and the assumptions with a yearly cash flow.
- **Results workbook (Excel)**: every number unrounded, for whoever checks the maths.
- **Project file (.json)**: your whole setup in one file. Open it from Home next time to carry on where you left off.
- The **owner memo** is shown below with a copy box. Switch on **Polish the wording with Claude** if an API key is set; it may reword but never change numbers.

4 A five-minute demo

A script for presenting the tool. The numbers are from the illustrative Pune demo and will change if you edit the data.

#	Do this	Say this
1	Home: click **Try the Pune demo**.	Developers pick green credits on upfront price. We cost them over the building's whole life instead.
2	1 Set up: **Import a vendor quote**, choose the SunRise solar quote, **Extract fields**, review, **Add to library**.	The quote is read automatically, but a person always checks it before it is used.
3	2 Recommendation: read the answer and the two route cards.	Both routes reach Gold. The recommended one costs Rs 1.10 Cr more upfront but is Rs 82.3 L better over 25 years.
4	Drag the **Quick what-if** sliders; open **Compare measures**.	Even with slower price growth it still wins. The cheaper-to-buy air-conditioning is not the cheaper-to-own one.
5	Switch the sidebar target to **Platinum**, then back.	The tool shows what the next level would cost over the building's life.
6	3 Can we trust it?: read the win rate and the tipping points.	The recommendation wins in 97% of 3,000 simulated futures.
7	4 Report: download the PDF and open it.	One document the owner, the IGBC consultant and the tender team can all use.

5 Using your own project

- **Start from the demo** and overwrite it: it is quicker than starting blank, and every field already has a sensible value.
- **Building & money first.** Enter areas, yearly energy and water use, today's tariff and its yearly growth, and your discount rate. Percentages are typed as %, e.g. 8 for 8%.
- **Then the measures.** Costs are **extra over a code-minimum baseline**: a VRF row holds the VRF price minus the split ACs it replaces. Energy saved is **kWh per year**. Solar rows use kWp instead of kWh; the tool works out the energy from the solar yield.
- **Exclusive group** marks alternatives: measures sharing a group (e.g. HVAC, PV) are never chosen together. Leave it blank for independent measures.
- **Replacement parts**: list anything re-bought during the study period (inverter, compressor) with its cost and life in years.
- **IGBC tables** shipped with the demo are illustrative. For real use, replace data/igbc_credits.csv and data/igbc_levels.csv with values from the official IGBC guide.
- **Save your work**: on 4 Report, download the **project file**. Next time, open it from Home. The app does not save automatically.

6 Reading the results

On screen	Textbook term	How to read it
Lifetime cost	Life-cycle cost (LCC)	Everything paid minus everything saved over the study period, in today's money. **Negative = it pays for itself.**
Lifetime cost after FAR value	Net LCC	Lifetime cost minus the value of the extra floor area the rating unlocks.
Cost per year	Equivalent annual cost (EAC)	Lifetime cost spread evenly per year; fair for comparing options with different lives.
Upfront cost	Capex	Extra money spent at the start, over the code-minimum baseline.

On screen	Textbook term	How to read it
Rs saved per Rs spent	Savings-to-investment ratio (SIR)	Above 1 means the measure is worth doing.
Payback	Simple payback	Years until savings have repaid the extra cost.
Wins in x% of futures	Monte Carlo probability	Above about 90% is robust; near 50% is a coin toss.
Tipping point	Break-even value	The value at which the two routes cost the same. Further from today = safer.
CO_2 avoided	Lifetime emissions avoided	kWh saved x grid emission factor, over the study period.

7 Problems and fixes

What you see	What to do
'No project loaded yet'	Go to Home and click Try the Pune demo, or open a saved project file.
A red list of errors after Apply	Each line names the field or row and the rule it breaks. Fix it and press Apply again.
'Target is not reachable'	Click Switch target, choose a lower level in the sidebar, or add measures that earn more points.
Results look too good or too bad	Check that measure costs are extra over baseline and that energy is per year, not per month.
The quote reader got a field wrong	It only drafts. Correct the field in the review form before Add to library.
'Could not open that file' on Home	Only project files saved from the Report page can be opened.
The page is slow the first time	It is searching every combination and simulating thousands of futures; later clicks reuse the results.
Browser didn't open	Open http://localhost:8501 yourself while the terminal is running.